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FIXING DEVICE

Abstract

A fixing device includes a heating member, a pressure member, a first side frame, a second side frame, a connection frame, an external connector, a first temperature sensor, a second temperature sensor, an intermediate circuit board, a first cable, a second cable, and a third cable. The fixing device is installable into a main housing of an image forming apparatus. The heating member and the pressure member are configured to nip a sheet therebetween. The connection frame connects the first side frame and the second side frame. The intermediate circuit board is configured to relay signals from the first temperature sensor and the second temperature sensor to the external connector. The first cable connects the first temperature sensor and the intermediate circuit board. The second cable connects the second temperature sensor and the intermediate circuit board. The third cable connects the intermediate circuit board and the external connector.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS
[0001] This application claims priority from Japanese Patent Application Nos. 2024-029977 and 2024-029982 filed on Feb. 29, 2024. The entire contents of the priority applications are incorporated herein by reference.
BACKGROUND ART
[0002] A fixing device installable into and removable from a main housing of an image forming apparatus is known in the art. Such a fixing device includes a heating member, side frames, an external connector, a plurality of sensors and an intermediate circuit board. The heating member includes a heating source. The side frames are located on both sides of the heating member and are connected by an exterior member made of plastic. The external connector transmits signals to the image forming apparatus. The signals from the plurality of sensors are relayed to the external connector by the intermediate circuit board. The intermediate circuit board is located frontward of the heating member and adjacent to the heating member and to the external connector. The intermediate circuit board is held by a retainer fixed to the exterior member.
SUMMARY
[0003] When an intermediate circuit board is located adjacent to an external connector, the cables connecting sensors to the intermediate circuit board may become longer depending on the positions of the sensors, making wiring complicated.
[0004] It would be desirable to make the cables connecting the sensors to the intermediate circuit board shorter and simplify the wiring.
[0005] In one aspect, the fixing device comprises a heating member, a pressure member, a first side frame, a second side frame, a connection frame, an external connector, a first temperature sensor, a second temperature sensor, an intermediate circuit board, a first cable, a second cable and a third cable. The fixing device is installable into a main housing of an image forming apparatus.
[0006] The heating member extends in a longitudinal direction and comprises a heater.
[0007] The pressure member extends in the longitudinal direction and nips a sheet in combination with the heating member.
[0008] The first side frame is located on one side in the longitudinal direction with respect to the heating member.
[0009] The second side frame is located on the other side in the longitudinal direction with respect to the heating member.
[0010] The connection frame extends in the longitudinal direction and connects the first side frame and the second side frame.
[0011] The external connector is connectable to the main housing.
[0012] The first temperature sensor detects a temperature of the heating member.
[0013] The second temperature sensor detects a temperature of the heating member.
[0014] The intermediate circuit board relays a signal from the first temperature sensor to the external connector and relays a signal from the second temperature sensor to the external connector.
[0015] The first cable connects the first temperature sensor and the intermediate circuit board.
[0016] The second cable connects the second temperature sensor and the intermediate circuit board.
[0017] The third cable connects the intermediate circuit board and the external connector.
[0018] Since the wiring of the sensors is brought together at the intermediate circuit board and then

connected to the external connector, wiring can be simplified if the sensors are located on the one side in the longitudinal direction with respect to a longitudinal center of the heating member.
[0019] The external connector may be located on the other side in the longitudinal direction with respect to the longitudinal center of the heating member, and a longitudinal center of the intermediate circuit board may be located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member.
[0020] Since the longitudinal center of the intermediate circuit board is located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member, the cables connecting the sensors and the intermediate circuit board can be made shorter if the sensors are located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member.
[0021] An edge of the intermediate circuit board on the other side in the longitudinal direction may be located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member.
[0022] Since the edge of the intermediate circuit board on the other side in the longitudinal direction is located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member, the cables connecting the sensors and the intermediate circuit board can be made shorter if the sensors are located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member.
[0023] An edge of the intermediate circuit board on the one side in the longitudinal direction may be closer, than an edge of the heating member on the one side in the longitudinal direction, to the longitudinal center of the heating member.
[0024] Since the edge of the intermediate circuit board on the one side in the longitudinal direction is closer, than an edge of the heating member on the one side in the longitudinal direction, to the longitudinal center of the heating member, i.e., since the intermediate circuit board is located within a longitudinal area in which the heating member is located, interference of the intermediate circuit board with a structure on an end of the fixing device on the one side in the longitudinal direction can be restrained.
[0025] The first temperature sensor may be located within a longitudinal area through which a sheet having a minimum width fixable in the fixing device passes, and the second temperature sensor may be located outside a longitudinal area through which a sheet having a maximum width fixable in the fixing device passes.
[0026] The first temperature sensor may be located between one end and the other end of the intermediate circuit board in the longitudinal direction, and the second temperature sensor may be located on the one side in the longitudinal direction with respect to an edge of the intermediate circuit board on the one side in the longitudinal direction.
[0027] Further, in order to increase the rigidity of the fixing device, it would be desirable to connect the side frames of the fixing device by a metal frame. However, if the intermediate circuit board is disposed on the metal frame, the intermediate circuit board would be susceptible to the heat from the heating element.
[0028] Thus, it would be desirable to increase the rigidity of the fixing device and to make the intermediate circuit board less susceptible to the heat from the heating member.
[0029] In light of the above, the connection frame may be made of metal and located between the intermediate circuit board and the heating member, and a first frame made of plastic may be provided between the intermediate circuit board and the connection frame.
[0030] By connecting the pair of side frames by a connection frame made of metal, the rigidity of the fixing device can be increased. Further, even if the connection frame is located between the intermediate circuit board and the heating member, the intermediate circuit board is less susceptible to the heat from the heating member because the first frame made of plastic is provided between the intermediate circuit board and the connection frame.

[0031] The intermediate circuit board may be located above the heating member.
[0032] Since the intermediate circuit board is located above the heating member, the dimension of the fixing device in the front-rear direction can be made smaller than that of a fixing device of an alternative configuration in which the intermediate circuit board is located frontward or rearward of the heating member.
[0033] The first temperature sensor may be located above the heating member.
[0034] The first temperature sensor may be held by the connection frame.
[0035] The first frame may comprise a base wall opposed to the intermediate circuit board, and a support wall extending from the base wall toward the intermediate circuit board to support the intermediate circuit board with a space being formed between the intermediate circuit board and the base wall.
[0036] Since the support wall supports the intermediate circuit board with a space being formed between the intermediate circuit board and the base wall, a layer of air can block the heat of the heating member from the intermediate circuit board.
[0037] The first frame may hold the third cable.
[0038] Since the first frame supports the third cable, the third cable can be protected from contact with the connection frame.
[0039] A second frame made of plastic may be provided between the first frame and the connection frame.
[0040] Since a second frame made of plastic is provided between the first frame and the connection frame, the heat of the heating member can be doubly blocked from the intermediate circuit board.
[0041] The fixing device may further comprise a first power cable that connects the external connector and one end of the heater on the one side in the longitudinal direction. The external connector may be located on the other side in the longitudinal direction with respect to a longitudinal center of the heating member. The first power cable may extend between the first frame and the second frame.
[0042] Since the first power cable extends between the first frame and the second frame, the power cable can be protected from contact with the connection frame.
[0043] The external connector may be located outward of the second side frame in the longitudinal direction.
[0044] Since the external connector is located outward of the second side frame in the longitudinal direction, interference of the external connector with the sheet can be restrained.
[0045] The fixing device may further comprise a connector holder that holds the external connector. The connector holder may extend from one side to the other side of the second side frame in the longitudinal direction and include a cable holder that holds the third cable.
[0046] Since the connector holder extends from one side to the other side of the second side frame in the longitudinal direction and holds the third cable, the third cable can be restrained from contacting the second side frame.
[0047] The fixing device may further comprise a cover that covers the intermediate circuit board.
[0048] Since the fixing device comprises the cover, the intermediate circuit board can be protected.
[0049] The cover may include a vent hole located above the intermediate circuit board.
[0050] Since the cover includes a vent hole located above the intermediate circuit board, air in the fixing device can be discharged through the vent hole.
[0051] The fixing device may further comprise a sheet sensor located on the one side in the longitudinal direction with respect to a longitudinal center of the heating member and configured to detect a sheet that has passed through between the heating member and the pressure member, and a fourth cable that connects the sheet sensor and the intermediate circuit board.
[0052] Since the sheet sensor is located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member, the fourth cable connecting the sheet sensor and the intermediate circuit board can be made shorter.

[0053] The fixing device may further comprise a pressure changing mechanism that comprises a cam and is configured to change a nip pressure between the heating member and the pressure member, a cam sensor located on the one side in the longitudinal direction with respect to a longitudinal center of the heating member and configured to detect a phase of the cam, and a fifth cable that connects the cam sensor and the intermediate circuit board.
[0054] Since the cam sensor is located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member, the fifth cable that connects the cam sensor and the intermediate circuit board can be made shorter.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0055] The above aspects, other advantages and further features will become more apparent by describing in detail illustrative, non-limiting embodiments thereof with reference to the accompanying drawings, in which:
[0056] FIG. **1** is a cross sectional view of a fixing device showing a cam located in a first phase.
[0057] FIG. **2** is a cross sectional view of the fixing device showing the cam located in a second phase.
[0058] FIG. **3** is a perspective view of a fixing frame.
[0059] FIG. **4** is a perspective view showing an arrangement of an intermediate circuit board, sensors, cables and a fixing connector relative to the fixing frame.
[0060] FIG. **5** is a perspective view of the fixing frame shown in FIG. **4** as viewed in a direction of installation of the fixing device.
[0061] FIG. **6** is an illustration showing wiring between a heater, the sensors, the intermediate circuit board and the fixing connector.
[0062] FIG. **7** is a perspective view showing a rear guide placed on the fixing frame shown in FIG. **4** and showing the intermediate circuit board by a dashed-double dotted line.
[0063] FIG. **8** is a perspective view showing a first cable holder placed on the rear guide shown in FIG. **7**.
[0064] FIG. **9** is a partial enlarged view of an end on the other side in the longitudinal direction of the fixing device shown in FIG. **8**.
[0065] FIG. **10** is a sectional view taken along line A-A of FIG. **8**.
[0066] FIG. **11** is an illustration showing an end of the fixing device on one side in the longitudinal direction as viewed upward from below.
[0067] FIG. **12** is a perspective view showing a cover placed over the intermediate circuit board shown in FIG. **8**.
[0068] FIG. **13** is a plan view corresponding to FIG. **7**.
[0069] FIG. **14** is a schematic view showing engagement of gears and connection of connectors in a state where the fixing device is installed in the main housing of the image forming apparatus.
DESCRIPTION
[0070] An embodiment of the present disclosure will be described in detail referring to the drawings where appropriate.
[0071] As shown in FIG. **1**, a fixing device **1** is a device for fixing a toner image onto a sheet S. The fixing device **1** is installable into a main housing M of an image forming apparatus P such as a printer (see FIG. **14**). The fixing device **1** comprises a heating roller **2** as an example of a heating member, a pressure roller **3** as an example of a pressure member, a fixing frame **4**, an arm **5**, a spring **6** and a cam **7**.
[0072] The heating roller **2** extends in a longitudinal direction and rotates about a rotation axis X. The heating roller **2** receives a driving force from a motor (not shown) provided in the image

forming apparatus P and rotates. The heating roller **2** comprises a roller **2**A and a heater H. In this embodiment, the heating roller **2** comprises two heaters H**1**, H**2**. The roller **2**A is a metal tube heated by the heater H. The heater H is located inside the roller **2**A. The heater H is, for example, a halogen heater.
[0073] A central portion CL**1** of the heating roller **2** in the longitudinal direction (longitudinal center CL**1** of the heating roller **2**) is a central portion of a longitudinal area through which a sheet S.sub.max having a maximum width fixable in the fixing device **1** passes.
[0074] In the following description, the longitudinal direction in which the heating roller **2** extends is simply referred to as “longitudinal direction”. The longitudinal direction is also the direction in which the rotation axis X of the heating roller **2** extends. One side in the longitudinal direction corresponds to the right side of the fixing device **1**, and the other side in the longitudinal direction corresponds to the left side of the fixing device **1** as viewed in the direction in which the fixing device **1** is installed into the main housing M.
[0075] The direction in which the fixing device **1** is installed into the main housing M is referred to as “installation direction” and the direction in which the fixing device **1** is removed from the main housing M is referred to as “removal direction”. The installation direction and the removal direction respectively correspond to frontward and rearward of the main housing M. The installation direction and the removal direction are collectively referred to as installation-removal direction. The front-rear direction and the up-down direction of the fixing device correspond to the front-rear direction and the up-down direction of the main housing M in a state where the fixing device **1** is installed in the main housing M. The sheet S is conveyed from the front toward the rear of the fixing device **1** and passes through the fixing device **1**. The sheet S that has passed through the fixing device **1** is conveyed upward with respect to the fixing device.
[0076] The pressure roller **3** extends in the longitudinal direction. The pressure roller **3** rotates in accordance with the heating roller **2** and nips the sheet S with the heating roller **2**. The pressure roller **3** comprises a shaft **3**A and a roller body **3**B. The shaft is, for example, made of metal. The roller body **3**B is, for example, made of rubber and covers an outer periphery of the shaft **3**A.
[0077] As shown in FIG. **3**, the fixing frame **4** comprises a first side frame **4**A, a second side frame **4**B and a connection frame **4**C. The first side frame **4**A is located on the one side in the longitudinal direction with respect to the heating roller **2** and the pressure roller **3**, and the second side frame **4**B is located on the other side in the longitudinal direction with respect to the heating roller **2** and pressure roller **3**. The first side frame **4**A and the second side frame **4**B support the heating roller **2** in a manner that allows the heating roller **2** to rotate.
[0078] The connection frame **4**C is a metal plate extending in the longitudinal direction. The connection frame **4**C connects the first side frame **4**A and the second side frame **4**B. The first side frame **4**A, the second side frame **4**B and the connection frame **4**C may be formed integrally as one part.
[0079] The cam **7** includes a cam **7**A and a cam **7**B. The cam **7**A is located in the vicinity of the first side plate **4**A. The cam **7**B is located in the vicinity of the second side plate **4**B. The cam **7**A and the cam **7**B are connected by a cam shaft **8**. The cam shaft **8** is rotatably supported by the first side frame **4**A and the second side frame **4**B.
[0080] Referring back to FIG. **1**, the arm **6**, the spring **6** and the cam **7** form a pressure changing mechanism for changing a nip pressure between the heating roller **2** and the pressure roller **3**. The pressure changing mechanism is capable of changing the nip pressure by moving at least one of the heating roller **2** and the pressure roller **3** relative to the other of the heating roller **2** and the pressure roller **3**. In the present embodiment, the pressure changing mechanism changes the nip pressure by moving the pressure roller **3** relative to the heating roller **2**.
[0081] The arm **5** has a first end **5**A, a second end **5**B, a first portion **5**C and a second portion **5**D. The arm **5** is rotatable about an arm axis **1**X.
[0082] The first end **5**A is rotatably supported by the first side frame **4**A via a shaft **5**E. The second

end **5**B comprises a cam follower **5**F. The cam follower **5**F is contactable with the cam **7**.
[0083] The first portion **5**C and the second portion **5**D are located between the first end **5**A and the second end **5**B. The first portion **5**C supports the pressure roller **3**. The second portion **5**D is a portion to which the spring **6** is connected. One end of the spring **6** is hooked on the second portion **5**D and the other end of the spring **6** is hooked on the first side frame **4**A. The spring **6** causes the pressure roller **3** to be pressed against the heating roller **2** via the arm **5**.
[0084] The cam **7**A is rotatably supported by the first side frame **4**A via the cam shaft **8** and is rotatable about a cam axis **2**X. Rotation of the cam **7**A causes the pressure roller **3** to move to change the nip pressure for nipping the sheet S. The cam **7** is rotatable between a first phase shown in FIG. **1** and a second phase shown in FIG. **2**.
[0085] The first phase is a phase in which the nip pressure assumes a first nip pressure. The second phase is a phase in which the nip pressure assumes a second nip pressure lower than the first nip pressure. The cam **7**A is rotatable from the first phase shown in FIG. **1** to the second phase shown in FIG. **2** by rotating in the counterclockwise direction.
[0086] The arm **5**, the spring **6** and the cam **7**B provided on the second side frame **4**B have structures symmetrical to the arm **5**, the spring **6** and the cam **7**A provided on the first side frame **4**A. Thus, a detailed description of the arm **5**, the spring **6** and the cam **7**B provided on the second side frame **4**B will be omitted.
[0087] As shown in FIG. **3**, the fixing device **1** further comprises a cam gear **9** and a fixing gear **10**. The cam gear **9** is fixed to an end of the cam shaft **8** on the one side in the longitudinal direction.
[0088] The cam gear **9** is connected to the cam **7** via the cam shaft **8** and transmits a driving force to the cam **7**. The cam gear **9** includes gear teeth **9**A and a flange **9**B extending from the vicinity of the roots of the gear teeth **9**A toward the one side in the longitudinal direction. The flange **9**B has a notch **9**C for detecting the phase of the cam **7**.
[0089] The fixing gear **10** includes gear teeth **10**A. The fixing gear **10** is fixed to an end of the roller **2**A on the one side in the longitudinal direction. The fixing gear **10** is coaxial with the heating roller **2** and rotates together with the heating roller **2** about the rotation axis X. The fixing gear **10** transmits a driving force to the heating roller **2**.
[0090] As shown in FIGS. **4** and **5**, the fixing device **1** further comprises a first temperature sensor S**1**, a second temperature sensor S**2**, a sheet sensor S**3**, a cam sensor S**4**, a fixing connector **11** as an external connector and an intermediate circuit board **12**.
[0091] The first temperature sensor S**1** detects a temperature of the heating roller **2**. The first temperature sensor S**1** is spaced apart from the heating roller **2**. The first temperature sensor S**1** is retained on the connection frame **4**C by screws or the like. The first temperature sensor S**1** is located above the heating roller **2**. The first temperature sensor S**1** is, for example, a thermistor.
[0092] The second temperature sensor S**2** detects a temperature of the heating roller **2**. The second temperature sensor S**2** is in contact with the heating roller **2**. The second temperature sensor S**2** is retained on the connection frame **4**C by screws or the like. The second temperature sensor S**2** is, for example, a thermistor.
[0093] The sheet sensor S**3** detects a sheet S that has passed through between the heating roller **2** and the pressure roller **3**. The sheet sensor S**3** comprises a rotation member **14** and an optical sensor **15**. The rotation member **14** rotates upon contact with the sheet S.
[0094] The optical sensor **15** comprises a light emitting element that emits light and a light receiving element that receives light from the light emitting element. The optical sensor **15** emits light from the light emitting element along the longitudinal direction.
[0095] The rotation member **14** comprises a first lever **14**A, a second lever **14**B and a shaft **14**C. The first lever **14**A is contactable with the sheet S. The second lever **14**B can block light of the optical sensor **15**. The shaft **14**C connects the first lever **14**A and the second lever **14**B. The rotation member **14** is rotatable about the shaft **14**C.
[0096] In a state where the rotation member **14** is not in contact with a sheet S as shown in FIGS. **4**

and **5**, light of the optical sensor **15** is blocked by the second lever **14**B of the rotation member **14**. The rotation member **14** rotates when the first lever **14**A is pushed by a sheet S. As a result, the second lever **14**B moves out of the light path of the optical sensor **15**. Thus, it can be determined that a sheet is not passing through between the heating roller **2** and the pressure roller **3** if the light receiving element of the optical sensor **15** is not receiving light, and that a sheet is passing through between the heating roller **2** and the pressure roller **3** if the light receiving element of the optical sensor **15** is receiving light.

[0097] The cam sensor S**4** detects the phase of the cam **7**. The cam sensor S**4** comprises an optical sensor **16**.

[0098] The optical sensor **16** comprises a light emitting element that emits light and a light receiving element that receives light from the light emitting element. The optical sensor **16** emits light from the light emitting element along the up-down direction.

[0099] In a state where the cam **7** is located in the first phase as shown in FIG. **4**, light of the optical sensor **16** is blocked by the flange **9**B of the cam gear **9**. After the cam **7** rotates to the second phase, the light receiving portion of the optical sensor **16** is able to receive light through the notch **9**C formed in the flange **9**B of the cam gear **9**. Thus, it can be determined that the cam **7** is located in the second phase if the light receiving element of the optical sensor **16** is receiving light, and that the cam is located in a phase other than the second phase if the light receiving element of the optical sensor **16** is not receiving light.

[0100] The fixing connector **11** is connectable to the main housing M in a state where the fixing device **1** is installed in the main housing M. In more detail, the fixing connector **11** is connectable to a housing connector **18** (see FIG. **14**) provided at the main housing M in a state where the fixing device **1** is installed in the main housing M. The fixing connector **11** is located outside of the first side frame **4**A and the second side frame **4**B in the longitudinal direction. In more detail, the fixing connector **11** is located outside of the second side frame **4**B.

[0101] The intermediate circuit board **12** relays signals from the first temperature sensor S**1**, the second temperature sensor S**2**, the sheet sensor S**3** and the cam sensor S**4** to the fixing connector **11**. The intermediate circuit board **12** comprises a second temperature sensor connector CN**1**, a sheet sensor connector CN**2**, a cam sensor connector CN**3**, a first temperature sensor connector CN**4**, and a relay connector CN**5**. The second temperature sensor connector CN**1**, the sheet sensor connector CN**2**, the cam sensor connector CN**3**, the first temperature sensor connector CN**4**, and the relay connector CN**5** are internal connectors.

[0102] As shown in FIG. **6**, the intermediate circuit board **12** is connected to the first temperature sensor S**1** by a first cable C**1**. The intermediate circuit board **12** is connected to the second temperature sensor S**2** by a second cable C**2**. The intermediate circuit board **12** is connected to the fixing connector **11** by a third cable C**3**. Further, the intermediate circuit board **12** is connected to the sheet sensor S**3** by a fourth cable C**4** and to the cam sensor S**4** by a fifth cable C**5**.

[0103] The heaters H**1**, H**2** of the heating roller **2** are connected to the fixing connector **11** at the one side in the longitudinal direction by a first power cable PC**1**. In more detail, the heaters H**1**, H**2** are connected to the fixing connector **11** by the power cable PC**1** through a thermostat TM. The heaters H**1**, H**2** are connected to the fixing connector **11** at the other side in the longitudinal direction by second power cables PC**2**, PC**2**. The thermostat TM shuts down power to the heaters H**1**, H**2** when the temperature of the heating roller **2** exceeds a controllable range and becomes overheated.

[0104] In FIG. **6**, the fixing device **1** is installed in the main housing M. In a state where the fixing device **1** is installed in the main housing M, the fixing connector **11** is connected to the housing connector **18**. When the fixing connector **11** and the housing connector **18** are connected, temperature information detected by the temperature sensors S**1**, S**2**, sheet information detected by the sheet sensor S**3** and the phase of the cam **7** detected by the cam sensor S**4** can be transmitted to a controller MC of the image forming apparatus P. Further, when the fixing connector **11** and the housing connector **18** are connected, power can be supplied to the heater H from a heater drive unit HC of the main housing M. The heater drive unit HC is controlled by the controller MC and supplies power to the heater H based on the temperature information detected by the temperature sensors S**1**, S**2**.

[0105] As shown in FIGS. **7** to **9**, the fixing device **1** further comprises a first cable holder **20** as a first frame and a rear guide **30** as a second frame.

[0106] As shown in FIG. **7**, the rear guide **30** is a member that guides a sheet S ejected through between the heating roller **2** and the pressure roller **3**. The rear guide **30** is made of plastic. The rear guide **30** includes a guide body **31** and a guide portion **32** extending upward from the guide body **31**.

[0107] The guide body **31** extends in the longitudinal direction and in the installation direction to cover the connection frame **4**C. The guide body **31** guides the first power cable PC**1** from an end of the heater H on the one side in the longitudinal direction to the fixing connector **11**. The guide body **31** includes a first retainer **33**, a second retainer **34**, a third retainer **35**, a fourth retainer **36**, a fifth retainer **37**, a sixth retainer **38**, and a first guide wall **39**.

[0108] The first retainer **33** and the second retainer **34** hold the first power cable PC**1**. A plurality of the first retainers **33** are provided. In more detail, two first retainers **33** are provided on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2** and one first retainer **33** is provided on the other side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**. Each first retainer **33** includes a pair of protrusions facing each other in the installation direction and protruding upward from the guide body **31**. The first retainer **33** holds the first power cable PC**1** between the pair of protrusions. The first retainer **33** restricts movement of the first power cable PC**1** in the installation-removal direction.

[0109] The second retainer **34** is located between the fixing connector **11** and the first retainer **33** located on the other side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**. The second retainer **34** is an L-shaped element protruding upward from the guide body **31**. The second retainer **34** includes an extending portion **34**A and a hook portion **34**B. The extending portion **34**A extends upward from the guide body **31**. The hook portion **34**B extends in the removal direction from an upper end of the extending portion **34**A. The second retainer **34** restricts movement of the first power cable PC**1** in the upward direction and in the installation direction.

[0110] The third retainer **35**, the fourth retainer **36** and the fifth retainer **37** hold the fourth cable C**4** and the fifth cable C**5**. The third retainer **35** is located at a frontward side of an end of the guide body **31** on the one side in the longitudinal direction. The third retainer **35** includes a pair of protrusions protruding from the guide body **31** in the installation direction. The third retainer **35** holds the fourth cable C**4** and the fifth cable C**5** between the pair of protrusions. The third retainer **35** restricts movement of the fourth cable C**4** and the fifth cable C**5** in the longitudinal direction.

[0111] The fourth retainer **36** is located upward and on the other side in the longitudinal direction with respect to the third retainer **35**. The fourth retainer **36** is an L-shaped element protruding upward from the guide body **31**. The fourth retainer **36** includes an extending portion **36**A and a hook portion **36**B. The extending portion **36**A extends upward from the guide body **31**. The hook portion **36**B extends in the installation direction from an upper end of the extending portion **36**A. The fourth retainer **36** restricts movement of the fourth cable C**4** and the fifth cable C**5** in the upward direction.

[0112] The fifth retainer **37** is located on the other side in the longitudinal direction with respect to the fourth retainer **36**. The fifth retainer **37** is an L-shaped element protruding from the guide body **31** in the installation direction. The fifth retainer **37** includes an extending portion **37**A and a hook portion **37**B. The extending portion **37**A extends in the installation direction from the guide body **31**. The hook portion **37**B extends downward from a front end of the extending portion **37**A. The fifth retainer **37** restricts movement of the fourth cable C**4** and the fifth cable C**5** in the upward

direction and in the installation-removal direction.
[0113] The sixth retainer **38** holds the first cable C**1** and the third cable C**3**. The sixth retainer **38** is located on the other side in the longitudinal direction with respect to the fifth retainer **37**. The sixth retainer **38** is an L-shaped element protruding from the guide body **31** in the installation direction. The sixth retainer **38** includes an extending portion **38**A and a hook portion **38**B. The extending portion **38**A extends in the installation direction from the guide body **31**. The hook portion **38**B extends downward from a front end of the extending portion **38**A. The sixth retainer **38** restricts movement of the first cable C**1** and the third cable C**3** in the upward direction and in the installation-removal direction.
[0114] The first guide wall **39** guides the fourth cable C**4** and the fifth cable C**5** in the longitudinal direction between the fourth retainer **36** and the fifth retainer **37**.
[0115] As shown in FIG. **8**, the first cable holder **20** is a member that supports the intermediate circuit board **12**. The first cable holder **20** is made of plastic. The first cable holder **20** includes a base wall **21**, a lug **21**A, a support wall **22**, a seventh retainer **23**, an eighth retainer **24**, a ninth retainer **25**, a tenth retainer **26**, an eleventh retainer **27**, a second guide wall **28** and a third guide wall **29**.
[0116] The base wall **21** extends in the longitudinal direction and in the installation direction to cover a portion of the guide body **31** of the rear guide **30**. The base wall **21** is opposed to the intermediate circuit board **12**. The lug **21**A extends upward from the base wall **21** toward the intermediate circuit board **12**.
[0117] The support wall **22** extends upward from the base wall **21** toward the intermediate circuit board **12**. The intermediate circuit board **12** is supported on an upper end of the support wall **22** (see also FIG. **10**). An end of the intermediate circuit board **12** on the one side in the longitudinal direction is fixed to the first cable holder **20** by a screw or the like. An end of the intermediate circuit board **12** on the other side in the longitudinal direction is held on the first cable holder **20** by the lug **21**A.
[0118] The seventh retainer **23** and the eighth retainer **24** hold the second cable C**2**. The seventh retainer **23** is located on a rearward side of an end of the base wall **21** on the one side in the longitudinal direction. The seventh retainer **23** is an L-shaped element protruding upward from the base wall **21**. The seventh retainer **23** includes an extending portion **23**A and a hook portion **23**B. The extending portion **23**A extends upward from the base wall **21**. The hook portion **23**B extends in the removal direction from an upper end of the extending portion **23**A. The seventh retainer **23** restricts movement of the second cable C**2** in the upward direction and in the installation direction.
[0119] The eighth retainer **24** is located frontward and on the other side in the longitudinal direction with respect to the seventh retainer **23**. The eighth retainer **24** includes a pair of protrusions protruding upward from the base wall **21**. The eighth retainer **24** holds the second cable C**2** between the pair of protrusions. The eighth retainer **24** restricts movement of the second cable C**2** in the installation-removal direction.
[0120] The ninth retainer **25**, the tenth retainer **26** and the eleventh retainer **27** hold the third cable C**3**.
[0121] The ninth retainer **25** is located on the other side in the longitudinal direction with respect to the eighth retainer **24**. As shown in FIG. **9**, the ninth retainer **25** includes an extending portion **25**A and a hook portion **25**B. The extending portion **25**A extends upward from the base wall **21**. The hook portion **25**B extends in the removal direction from an upper end of the extending portion **25**A. The ninth retainer **25** restricts movement of the third cable C**3** in the upward direction and in the installation direction.
[0122] The tenth retainer **26** is located on the other side in the longitudinal direction with respect to the ninth retainer **25**. The tenth retainer **26** includes a first extending portion **26**A, a second extending portion **26**B and a hook portion **26**C. The first extending portion **26**A extends upward from the base wall **21**. The second extending portion **26**B extends in the installation direction from

the first extending portion **26**A. The hook portion **26**C extends downward from a forward end of the second extending portion **26**B. The tenth retainer **26** restricts movement of the third cable C**3** in the upward direction and in the installation-removal direction.

[0123] The eleventh retainer **27** is located on the other side in the longitudinal direction with respect to the tenth retainer **26**. The eleventh retainer **27** includes an extending portion **27**A and a hook portion **27**B. The extending portion **27**A extends upward from the base wall **21**. The hook portion **27**B extends in the removal direction from the extending portion **27**A. The eleventh retainer **27** restricts movement of the third cable C**3** in the upward direction and in the installation direction.

[0124] The second guide wall **28** is located between the ninth retainer **25** and the tenth retainer **26** and guides the third cable C**3** in the longitudinal direction.

[0125] The third guide wall **29** is located rearward of the second guide wall **28** and between the tenth retainer **26** and the eleventh retainer **27**. The third guide wall **29** guides the third cable C**3** in the longitudinal direction.

[0126] The fixing device **1** further comprises a connector holder **40**. The connector holder **40** is a member that holds the fixing connector **11**. The connector holder **40** includes a base portion **41**, a first guide hole **42**, a second guide hole **43** and a second cable holder **50**.

[0127] The base portion **41** is located around the fixing connector **11** to hold the fixing connector **11**. The first guide hole **42** and the second guide hole **43** are engaged with guide pins (not shown) of the housing connector **18** in a state where the fixing device **1** is installed in the main housing M.

[0128] The second cable holder **50** extends from one side to the other side of the second side frame **4**B. The second side frame **4**B has an edge **4**E facing a direction perpendicular to the longitudinal direction (upward in this embodiment). The second cable holder **50** extends between the edge **4**E of the second side frame **4**B, and the third cable C**3** and the first power cable PC**1**. The second cable holder **50** includes a holder body **51**, a twelfth retainer **52** and a thirteenth retainer **53**.

[0129] The twelfth retainer **52** holds the first power cable PC**1**. The twelfth retainer **52** is located on the other side in the longitudinal direction with respect to the second retainer **34** (see FIG. **7**). The twelfth retainer **52** extends toward the one side in the longitudinal direction from the holder body **51**. The twelfth retainer **52** has a groove **52**A that guides the first power cable PC**1**. The twelfth retainer **52** restricts movement of the first power cable PC**1** in the installation-removal direction.

[0130] The thirteenth retainer **53** holds the third cable C**3**. The thirteenth retainer **53** is located on the other side in the longitudinal direction with respect to the eleventh retainer **27**. The thirteenth retainer **53** is an L-shaped element protruding in the installation direction from the holder body **51**. The thirteenth retainer **53** includes an extending portion **53**A and a hook portion **53**B. The extending portion **53**A extends in the installation direction from the holder body **51**. The hook portion **23**B extends downward from a front end of the extending portion **53**A. The thirteenth retainer **53** restricts movement of the third cable C**3** in the upward direction and in the installation-removal direction.

[0131] As shown in FIG. **10**, the intermediate circuit board **12** is located above the heating roller **2**. The intermediate circuit board **12** is located on a side of the connection frame **4**C opposite to a side of the connection frame **4**C on which the heating roller **2** is located, i.e., the connection frame **4**C is located between the intermediate circuit board **12** and the heating roller **2**. The first cable holder **20** and the guide body **31** of the rear guide **30** are located between the intermediate circuit board **12** and the connection frame **4**C in this order from the top. That is, the guide body **31** of the rear guide **30** is located between the first cable holder **20** and the connection frame **4**C. The first power cable PC**1** extends between the first cable holder **20** and the guide body **31** of the rear guide **30**. The support wall **22** of the first cable holder **20** supports the intermediate circuit board **12** with a space SP being formed between the intermediate circuit board **12** and the base wall **21**.

[0132] FIG. **11** is an illustration showing an end of the fixing device on the one side in the longitudinal direction viewed upward from below.

[0133] The fixing device **1** further comprises a gear cover **60** that covers part of the gear teeth **10**A of the fixing gear **10**. The gear cover **60** covers the gear teeth **10**A on the bottom half of the fixing gear **10**.

[0134] The gear cover **60** includes a fourteenth retainer **61**, a fifteenth retainer **62** and a sixteenth retainer **63** (see FIG. **8**).

[0135] The fourteenth retainer **61** holds the fifth cable C**5**. The fourteenth retainer **61** is located below the cam sensor S**4**. The fourteenth retainer **61** is an L-shaped element extending in the removal direction from the gear cover **60**. The fourteenth retainer **61** includes an extending portion **61**A and a hook portion **61**B. The extending portion **61**A extends in the removal direction from the gear cover **60**. The hook portion **61**B extends toward the other side in the longitudinal direction from a rear end of the extending portion **61**A. The fourteenth retainer **61** restricts movement of the fifth cable C**5** in the installation-removal direction and in the longitudinal direction.

[0136] The fifteenth retainer **62** and the sixteenth retainer **63** hold the fourth cable C**4** and the fifth cable C**5**. The fifteenth retainer **62** is located frontward of the fourteenth retainer **61**. The fifteenth retainer **62** is a plate-shaped element that covers the fourth cable C**4** and the fifth cable C**5** from the one side in the longitudinal direction. The fifteenth retainer **62** restricts movement of the fourth cable **4** and the fifth cable C**5** in the longitudinal direction.

[0137] As shown in FIG. **8**, the sixteenth retainer **63** is located on the other side in the longitudinal direction with respect to the fifteenth retainer **62**. The sixteenth retainer **63** is an L-shaped element protruding toward the other side in the longitudinal direction from the gear cover **60**. The sixteenth retainer **63** includes an extending portion **63**A and a hook portion **63**B. The extending portion **63**A extends toward the other side in the longitudinal direction from the gear cover **60**. The hook portion **63**B extends downward from a front end of the extending portion **63**A. The sixteenth retainer **63** restricts movement of the fourth cable C**4** and the fifth cable C**5** in the upward direction and in the installation-removal direction.

[0138] Next, an example of a procedure for attaching the first power cable PC**1** to the fixing device **1** will be described referring to FIGS. **7** and **9**.

[0139] The first power cable PC**1** has a first portion on the one side in the longitudinal direction connected to the heater H and a second portion on the other side in the longitudinal direction connected to the fixing connector **11**. Upon attaching the first power cable PC**1** to the fixing device **1**, the first portion of the first power cable PC**1** is drawn upward from the heater H and then in the longitudinal direction along the gear cover **60**. The first portion of the first power cable PC**1** is then drawn through between each of the pairs of protrusions of the two first retainers **33** located on the one side in the longitudinal direction and is connected to the thermostat TM. Next, the second portion of the first power cable PC**1** is drawn rearward from the fixing connector **11** and then along the holder body **51** of the second cable holder **50** to place the second portion of the first power cable PC**1** in the groove **52**A of the twelfth retainer **52**. The second portion of the first power cable PC**1** is then hooked on the second retainer **34**, drawn through the pair of protrusions of the first retainer **33** located on the other side in the longitudinal direction and connected to the thermostat TM.

[0140] Next, an example of a procedure for attaching the cables C**1** to C**5** to the fixing device **1** will be described referring to FIGS. **8**, **9** and **11**.

[0141] As shown in FIG. **8**, upon attaching the first cable C**1** to the fixing device **1**, the first cable C**1** is drawn upward from the first temperature sensor S**1** and under the sixth retainer **38**, and is connected to the first temperature connector CN**4** of the intermediate circuit board **12**.

[0142] Upon attaching the second cable C**2** to the fixing device **1**, the second cable C**2** is drawn upward from the second temperature sensor S**2** and is hooked on the seventh retainer **23**. Subsequently, the second cable C**2** is drawn through between the pair of protrusions of the eighth retainer **24** and is connected to the second temperature sensor connector CN**1** of the intermediate circuit board **12**.

[0143] Upon attaching the third cable C**3** to the fixing device **1**, the third cable C**3** is drawn toward the one side in the longitudinal direction from the fixing connector **11** and around the holder body **51** of the second cable holder **50**. Subsequently, the third cable C**3** is drawn under the thirteenth retainer **53** and the eleventh retainer **27**, along the rear surface of the third guide wall **29**, under the tenth retainer **26** and the ninth retainer **25** and along the rear surface of the second guide wall **28**. The third cable C**3** is then drawn under the sixth retainer **38** and is connected to the relay connector CN**5** of the intermediate circuit board **12**.

[0144] Upon attaching the fourth cable C**4** to the fixing device **1**, the fourth cable C**4** is drawn upward from the sheet sensor S**3** and along an outer periphery of a portion of the gear cover **60** that covers the fixing gear **10** as shown in FIG. **11**. The fourth cable C**4** is then drawn through the fifteenth retainer **62** and the sixteenth retainer **63** as shown in FIG. **8** and then upward through between the pair of protrusions of the third retainer **35**. Subsequently, the fourth cable C**4** is drawn under the fourth retainer **36**, along the rear surface of the first guide wall **39** and under the fifth retainer **37**, and is connected to the sheet sensor connector CN**2** of the intermediate circuit board **12**.

[0145] As shown in FIG. **11**, the cam sensor S**4** is located above the heating roller **2**. Upon attaching the fifth cable C**5** to the fixing device **1**, the fifth cable C**5** is drawn downward from the cam sensor S**4** and hooked on the fourteenth retainer **61**. Next, the fifth cable C**5** is drawn along the outer periphery of the portion of the gear cover **60** that covers the fixing gear **10**, in such a manner that the fifth cable **5** extends under the heating roller **2**. Then, as shown in FIG. **8**, the fifth cable C**5** is drawn through the fifteenth retainer **62**, the sixteenth retainer **63**, the third retainer **35**, the fourth retainer **36** and the fifth retainer **37** in a similar manner as the fourth cable C**4**, and is connected to the cam sensor connector CN**3** of the intermediate circuit board **12**.

[0146] As shown in FIG. **12**, the fixing device **1** further comprises a cover **70** that covers the intermediate circuit board **12**. The cover **70** has a plurality of vent holes **71** located above the intermediate circuit board **12**.

[0147] The vent holes **71** allow air in the fixing device to be discharged.

[0148] As shown in FIG. **13**, the first temperature sensor S**1** is located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2** and detects a temperature of the heating roller **2** on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**. The first temperature sensor S**1** (a heat-sensitive element of the first temperature sensor S**1**) is located between one end and the other end of the intermediate circuit board **12** in the longitudinal direction. Further, the first temperature sensor S**1** is located within a longitudinal area through which a sheet S.sub.min having a minimum width fixable in the fixing device passes when the sheet S.sub.min is placed in a centered position.

[0149] The second temperature sensor S**2** is located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2** and detects a temperature of the heating roller **2** on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**. The second temperature sensor S**2** is located on the one side in the longitudinal direction with respect to an edge of the intermediate circuit board **12** on the one side in the longitudinal direction. Further, the second temperature sensor S**2** is located outside the longitudinal area through which the sheet S.sub.max having a maximum width fixable in the fixing device passes.

[0150] The fixing connector **11** is located on the other side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**.

[0151] The sheet sensor S**3** is located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**.

[0152] The cam sensor S**4** is located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**.

[0153] A longitudinal center CL**2** of the intermediate circuit board **12** is located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**. An edge of the intermediate circuit board **12** on the other side in the longitudinal direction is located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**. An edge of the intermediate circuit board **12** on the one side in the longitudinal direction is located closer, than an edge of the heating roller **2** on the one side in the longitudinal direction, to the longitudinal center CL**1** of the heating roller **2**.

[0154] The cam gear **9** is located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**.

[0155] In FIG. **14**, the fixing device **1** before installation into the main body M is shown by dashed-double dotted lines and the fixing device **1** after installation into the main body M is shown by solid lines. A fixing drive gear **80** and a cam drive gear **81** are further provided in the main body M of the image forming apparatus P. The fixing drive gear **80** rotates when it receives a driving force from a motor (not shown). The fixing device **1** is installable into the main housing M when a cover **90** provided on a rear side of the main housing M is open. An image forming unit **100** that forms a toner image on a sheet is located inside the main housing M. The image forming unit **100** supplies a sheet with a toner image formed thereon to the fixing device **1**.

[0156] In a state where the fixing device **1** is installed in the main housing M, the fixing gear **10** is meshed with the fixing drive gear **80** and rotates when it receives a driving force from the fixing drive gear **80**. In a state where the fixing device **1** is installed in the main housing M, the cam gear **9** is meshed with the cam drive gear **81** and rotates when it receives a driving force from the cam drive gear **81**.

[0157] According to the above, the following advantageous effects may be obtained by the present embodiment.

[0158] In the present embodiment, the first temperature sensor S**1**, the second temperature sensor S**2**, the sheet sensor S**3**, and the cam sensor S**4**, located on the one side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**, are respectively connected to the intermediate circuit board **12** by the first cable C**1**, the second cable C**2**, the fourth cable C**4** and the fifth cable C**5**. Further, the intermediate circuit board **12** is connected to the fixing connector **11**, located on the other side in the longitudinal direction with respect to the longitudinal center CL**1** of the heating roller **2**, by a third cable C**3**.

[0159] The cables C**1** to C**5** each include a plurality of signal lines. The number of signal lines included in the third cable C**3** connecting the intermediate circuit board **12** and the fixing connector **11** is less than the total number of signal lines included in the first cable C**1**, the second cable C**2**, the fourth cable C**4** and the fifth cable C**5** connecting the first temperature sensor S**1**, the second temperature sensor S**2**, the sheet sensor S**3**, and the cam sensor S**4** to the intermediate circuit board **12**. This is because ground wires and the like can be shared. As a result, the wiring connecting the first temperature sensor S**1**, the second temperature sensor S**2**, the sheet sensor S**3** and the cam sensor S**4** to the fixing connector **11** in the fixing device **1** can be simplified.

[0160] The intermediate circuit board **12** is located on the one side in the longitudinal direction with respect the longitudinal center CL**1** of to the heating roller **2**; thus, the length of the first cable C**1**, the second cable C**2**, the fourth cable C**4** and the fifth cable C**5** which connect the first temperature sensor S**1**, the second temperature sensor S**2**, the sheet sensor S**3** and the cam sensor S**4** to the intermediate circuit board **12** can be made shorter.

[0161] The intermediate circuit board **12** is located above the heating roller **2**; thus, the dimension of the fixing device **1** in the front-rear direction can be made smaller than that of a fixing device of an alternative configuration in which the intermediate circuit board **12** is located frontward or rearward of the heating roller **2**.

[0162] Since the fixing connector **11** is located outside the pair of side frames **4**A, **4**B in the longitudinal direction, interference of the fixing connector **11** with the sheet S can be restrained.

[0163] The fixing gear **10** and the cam gear **9** are located on the one side in the longitudinal direction with respect to the heating roller **2**, that is, on the side opposite to the fixing connector **11**

in the longitudinal direction; thus, the fixing device **1** may be downsized. In addition, interference of the fixing gear **10** and the cam gear **9** with the fixing connector **11** can be restrained.
[0164] Since the pair of side frames **4**A, **4**B of the fixing device **1** is connected by the metal connection frame **4**C, the rigidity of fixing device **1** may be increased.
[0165] The first cable holder **20** is provided between the intermediate circuit board **12** and the connection frame **4**C; thus, the intermediate circuit board **12** is less susceptible to the heat from the heating roller **2** even if the intermediate circuit board **12** is located on the side of the connection frame **4**C opposite to the side of the connection frame **4**C on which the heating roller **2** is located.
[0166] The first cable holder **20** includes a base wall **21** and a support wall **22**, and the support wall **22** supports the intermediate circuit board **12** with a space SP being formed between the intermediate circuit board **12** and the base wall **21**; thus, a layer of air formed by the space SP can block the heat of the heating roller **2** from the intermediate circuit board **12**.
[0167] The guide body **31** of the plastic rear guide **30** is provided between the first cable holder **20** and the connection frame **4**C; thus, the first cable holder **20** and the guide body **31** can doubly block the heat of the heating roller **2** from the intermediate circuit board **12**.
[0168] The fifth cable C**5** connecting the cam sensor S**4** and the intermediate circuit board **12** extends under the heating roller **2**; thus, the fifth cable C**5** is less susceptible to the heat from the heating roller **2**. In addition, since the fifth cable C**5** is held by the gear cover **60**, contact of the fixing gear **10** with the fifth cable C**5** can be restrained.
[0169] Since the first cable holder **20** holds the third cable C**3**, the third cable C**3** can be protected from contact with the connection frame **4**C.
[0170] The first power cable PC**1** extends between the first cable holder **20** and the guide body **31** of the rear guide **30**; thus, the first power cable PC**1** can be protected from contact with the connection frame **4**C.
[0171] The second cable holder **50** of the connector holder **40** extends between the edge **4**E of the second side frame **4**B facing the direction perpendicular to the longitudinal direction and the third cable C**3**; thus, the third cable C**3** can be restrained from contacting the second side frame **4**B.
[0172] The fixing device **1** is installable into and removable from the main housing M of the image forming apparatus P. Since the fixing device **1** comprises a cover **70** that covers the intermediate circuit board **12**, the intermediate circuit board **12** can be protected from contact with other components inside the main housing M and from contact by a person when the fixing device **1** is being installed into or removed from the main housing M.
[0173] Since the cover **70** has vent holes **71** located above the intermediate circuit board **12**, air heated by convection caused by a fan and other components located in the fixing device **1** can be discharged to the outside of the fixing device **1** through the vent holes **71**.
[0174] While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:
[0175] In the above-described embodiment, the fixing device **1** is installable into and removable from the main housing M of the image forming apparatus P; however, the fixing device may be fixed to the main housing M such that it is not removable.
[0176] Although the heating roller **2** is adopted as the heating member in the above-described embodiment, the heating member may be, for example, a cylindrical fixing belt slidably supported

by a guide.
[0177] In the above-described embodiment, the side frames **4**A, **4**B support the heating roller **2** in a manner that allows the heating roller **2** to rotate; however, the side frames **4**A, **4**B may support the pressure roller **3** in a manner that allows the pressure roller **3** to rotate.
[0178] In the above-described embodiment, the heating roller **2** is driven and the pressure roller **3** rotates in accordance with the heating roller **2**; however, the pressure roller **3** may be driven and the heating roller **2** may rotate in accordance with the pressure roller **3**. In this case, the fixing gear **10** is fixed to the pressure roller **3** and transmits a driving force to the pressure roller **3**.
[0179] In the above-described embodiment, the pressure changing mechanism changes the nip pressure by moving the pressure roller **3** relative to the heating roller **2**; however, the pressure changing mechanism may change the nip pressure by moving the heating roller **2** relative to the pressure roller **3**. Alternatively, the fixing device **1** may not include the pressure changing mechanism.
[0180] Although a halogen heater is given as an example of the heater H in the above-described embodiment, the heater H may be a resistive heating element or an IH (induction heating) heat source and the like. Here, the IH heat source is a heat source that does not heat up itself, but heats the roller or the belt by magnetic induction heating. Although the fixing device **1** comprises two heaters H**1**, H**2** in the above-described embodiment, the fixing device **1** may include one or more than two heaters H.
[0181] The elements described in the above embodiment and its modified examples may be implemented selectively and in combination.

Claims

1. A fixing device installable into a main housing of an image forming apparatus, comprising: a heating member extending in a longitudinal direction and comprising a heater; a pressure member extending in the longitudinal direction and configured to nip a sheet in combination with the heating member; a first side frame located on one side in the longitudinal direction with respect to the heating member; a second side frame located on another side in the longitudinal direction with respect to the heating member; a connection frame extending in the longitudinal direction and connecting the first side frame and the second side frame; an external connector connectable to the main housing; a first temperature sensor configured to detect a temperature of the heating member; a second temperature sensor configured to detect a temperature of the heating member; an intermediate circuit board configured to relay a signal from the first temperature sensor to the external connector and to relay a signal from the second temperature sensor to the external connector; a first cable configured to connect the first temperature sensor and the intermediate circuit board; a second cable configured to connect the second temperature sensor and the intermediate circuit board; and a third cable configured to connect the intermediate circuit board and the external connector.
2. The fixing device according to claim 1, wherein the external connector is located on the another side in the longitudinal direction with respect to a longitudinal center of the heating member, and a longitudinal center of the intermediate circuit board is located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member.
3. The fixing device according to claim 2, wherein an edge of the intermediate circuit board on the another side in the longitudinal direction is located on the one side in the longitudinal direction with respect to the longitudinal center of the heating member.
4. The fixing device according to claim 3, wherein an edge of the intermediate circuit board on the one side in the longitudinal direction is closer, than an edge of the heating member on the one side in the longitudinal direction, to the longitudinal center of the heating member.
5. The fixing device according to claim 2, wherein the first temperature sensor is located within a

longitudinal area through which a sheet having a minimum width fixable in the fixing device passes, and the second temperature sensor is located outside a longitudinal area through which a sheet having a maximum width fixable in the fixing device passes.
6. The fixing device according to claim 2, wherein the first temperature sensor is located between one end and another end of the intermediate circuit board in the longitudinal direction, and the second temperature sensor is located on the one side in the longitudinal direction with respect to an edge of the intermediate circuit board on the one side in the longitudinal direction.
7. The fixing device according to claim 1, wherein the connection frame is made of metal and located between the intermediate circuit board and the heating member, and a first frame made of plastic is provided between the intermediate circuit board and the connection frame.
8. The fixing device according to claim 7, wherein the intermediate circuit board is located above the heating member.
9. The fixing device according to claim 8, wherein the first temperature sensor is located above the heating member.
10. The fixing device according to claim 7, wherein the first temperature sensor is held by the connection frame.
11. The fixing device according to claim 7, wherein the first frame comprises: a base wall opposed to the intermediate circuit board; and a support wall extending from the base wall toward the intermediate circuit board to support the intermediate circuit board with a space being formed between the intermediate circuit board and the base wall.
12. The fixing device according to claim 7, wherein the first frame is configured to hold the third cable.
13. The fixing device according to claim 7, wherein a second frame made of plastic is provided between the first frame and the connection frame.
14. The fixing device according to claim 13, further comprising a first power cable configured to connect the external connector and one end of the heater on the one side in the longitudinal direction, wherein the external connector is located on the another side in the longitudinal direction with respect to a longitudinal center of the heating member, and the first power cable extends between the first frame and the second frame.
15. The fixing device according to claim 1, wherein the external connector is located outward of the second side frame in the longitudinal direction.
16. The fixing device according to claim 15, further comprising a connector holder configured to hold the external connector, wherein the connector holder extends from one side to another side of the second side frame in the longitudinal direction and includes a cable holder configured to hold the third cable.
17. The fixing device according to claim 1, further comprising a cover configured to cover the intermediate circuit board.
18. The fixing device according to claim 17, wherein the cover includes a vent hole located above the intermediate circuit board.
19. The fixing device according to claim 1, further comprising: a sheet sensor located on the one side in the longitudinal direction with respect to a longitudinal center of the heating member and configured to detect a sheet that has passed through between the heating member and the pressure member; and a fourth cable configured to connect the sheet sensor and the intermediate circuit board.
20. The fixing device according to claim 1, further comprising: a pressure changing mechanism comprising a cam and configured to change a nip pressure between the heating member and the pressure member; a cam sensor located on the one side in the longitudinal direction with respect to a longitudinal center of the heating member and configured to detect a phase of the cam; and a fifth cable configured to connect the cam sensor and the intermediate circuit board.